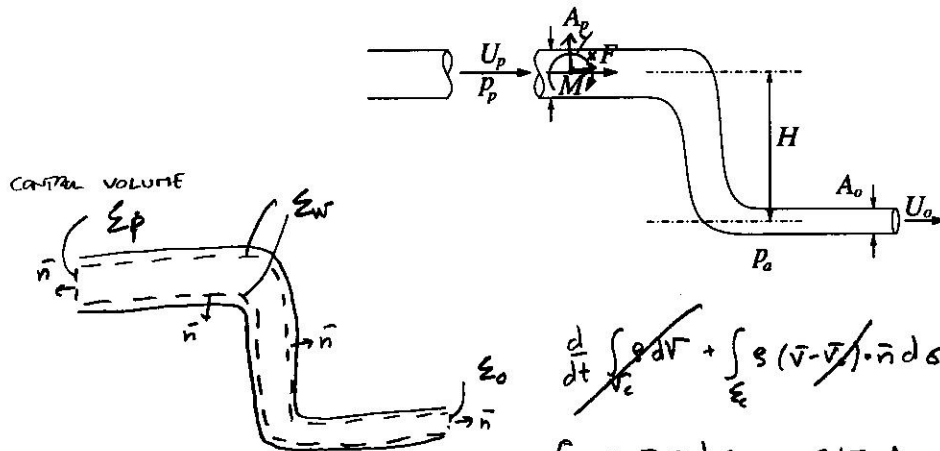


The double-elbow mouth piece of exit cross section A_o shown in the figure is to be installed at the end of a circular pipe of cross section A_p to redirect the outflow of water into the open atmosphere. Once the mouth piece is installed, the uniform values U_p and p_p are measured for the flow velocity and the pressure at the end of the pipe (i.e., just at the entrance of the mouth piece). Find the value of the uniform velocity at the exit section, U_o , as well as the force exerted on the mouth piece and its corresponding moment with respect to the entrance section. Neglect the effect of gravity.

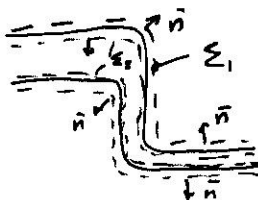


$$\frac{d}{dt} \int_{V_c} \rho dV + \int_{\Sigma} \rho (\vec{v} - \vec{v}_c) \cdot \vec{n} d\sigma = 0$$

$$\int_{\Sigma_p + \Sigma_o + \Sigma_w} \rho \vec{v} \cdot \vec{n} d\sigma = -\rho U_p A_p + \rho U_o A_o = 0 \Rightarrow \boxed{U_o = \frac{A_p}{A_o} U_p}$$

$\hookrightarrow \vec{v} \cdot \vec{n} = 0$

FORCE



$$\vec{F}_{MP} = - \int_{\Sigma_1} p_a \vec{n} d\sigma + \int_{\Sigma_1} \vec{z}' \cdot \vec{n} d\sigma - \int_{\Sigma_2} p \vec{n} d\sigma + \int_{\Sigma_2} \vec{z}' \cdot \vec{n} d\sigma \quad \text{② ALTERNATIVELY,}$$

$$= - \int_{\Sigma_2} (p - p_a) \vec{n} d\sigma + \int_{\Sigma_2} \vec{z}' \cdot \vec{n} d\sigma$$

$$\text{①}$$

$$- \int p_a \vec{n} d\sigma + \int p_a \vec{n} d\sigma + \int p_a \vec{n} d\sigma = - p_a \vec{e}_x (A_p - A_o)$$

$$\frac{d}{dt} \int_{V_c} \rho \vec{v} dV + \int_{\Sigma_c} \rho \vec{v} [(\vec{v} - \vec{v}_c) \cdot \vec{n}] d\sigma = - \int_{\Sigma_c} p \vec{n} d\sigma + \int_{\Sigma_c} \vec{z}' \cdot \vec{n} d\sigma + \int_{V_c} \rho \vec{f} dV$$

$$= - \rho U_p^2 A_p \vec{e}_x + \rho U_o^2 A_o \vec{e}_x$$

$$\int_{\Sigma_p + \Sigma_o + \Sigma_w} \vec{z}' \cdot \vec{n} d\sigma$$

$$- \int_{\Sigma_c} p \vec{n} d\sigma = - \int_{\Sigma_c} (p - p_a) \vec{n} d\sigma = (p_p - p_a) A_p \vec{e}_x - \int_{\Sigma_w} (p - p_a) \vec{n} d\sigma$$

COLLECTING TERMS YIELDS FINALLY

$$- \rho U_p A_p (U_p - U_o) \vec{e}_x = (p_p - p_a) A_p \vec{e}_x - \underbrace{\int_{\Sigma_w} (p - p_a) \vec{n} d\sigma}_{-\vec{F}_{MP}} + \int_{\Sigma_w} \vec{z}' \cdot \vec{n} d\sigma$$

TORQUE WITH RESPECT TO ENTRANCE

$$\vec{M}_{MP} = - \int_{\Sigma_1} (\vec{x} \wedge p_a \vec{n}) d\sigma + \int_{\Sigma_1} (\vec{x} \wedge \vec{z}' \cdot \vec{n}) d\sigma - \int_{\Sigma_2} \vec{x} \wedge p \vec{n} d\sigma + \int_{\Sigma_2} \vec{x} \wedge \vec{z}' \cdot \vec{n} d\sigma = - \int_{\Sigma_2} (p - p_a) \vec{x} \wedge \vec{n} d\sigma + \int_{\Sigma_2} \vec{x} \wedge \vec{z}' \cdot \vec{n} d\sigma$$

ANGULAR MOMENTUM CONSERVATION IN THE CONTROL VOLUME GIVES

$$\frac{d}{dt} \int_{V_c} (\vec{x} \wedge \rho \vec{v}) dV + \int_{\Sigma_c} (\vec{x} \wedge \rho \vec{v}) [(\vec{v} - \vec{v}_c) \cdot \vec{n}] d\sigma = - \int_{\Sigma_c} \vec{x} \wedge p \vec{n} d\sigma + \int_{\Sigma_c} \vec{x} \wedge \vec{z}' \cdot \vec{n} d\sigma + \int_{V_c} (\vec{x} \wedge \rho \vec{f}) dV$$

$$= - \int_{\Sigma_c} \vec{x} \wedge (p - p_a) \vec{n} d\sigma = - \int_{\Sigma_w} \vec{x} \wedge (p - p_a) \vec{n} d\sigma = \vec{M}_{MP}$$

$$\rho U_o^2 H A_o \vec{e}_z$$

$$\boxed{\vec{M}_{MP} = - \rho U_o^2 H A_o \vec{e}_z}$$